

	Orden	No Orden
1	✓	
2	✓	
3	✓	
4	✓	
5		✓
6		✓
7	✓	
8	✓	
9	✓	✓
10		✓

① $L = \{ \text{Palíndromos de } a, b \}$

$$S \rightarrow aSa \mid bSb \mid a \mid b \mid \lambda$$

② $L = \{ a^i b^j c^k \mid i=j \text{ o } j=k, i, j, k > 0 \}$

$$S \rightarrow A \mid B$$

$$A \rightarrow E \cdot F$$

$$E = aEb \mid ab$$

$$F = cF \mid c$$

$$A \rightarrow i=j$$

$$B \rightarrow GH$$

$$G = aG \mid a$$

$$H = bHc \mid bc$$

③ $\{ a^i b^i a^j b^j \mid i, j > 0 \}$

$$S \rightarrow A \mid B$$

$$A \rightarrow aAb \mid ab$$

$$B \rightarrow aAb \mid ab$$

④ $L = \{ a^i b^j c^k \mid k = i + 2j \}$

$$S \rightarrow \lambda \mid A$$

$$A \rightarrow aAc \mid B \mid \lambda$$

$$B \rightarrow bBcc \mid bcc \mid \lambda$$

⑤ $L = \{ (a+b)^n \mid N(b) \geq N(a) \}$

$$S \rightarrow SaSaSbS \mid SaSbSaS \mid SbSaSaS \mid \lambda$$

⑥ $L = (0+1)^* \mid N(0) \geq N(1) + 1$

$$S \rightarrow A0A$$

$$A \rightarrow A0A1A \mid A1A0A$$

**Nota: Aquí no valen las combinaciones
contraejemplo:*

bbaaaaaabb

⑦ $L = \{ a^i (b+c)^k \mid k > i \}$

$$S \rightarrow aSX \mid X$$

$$X \rightarrow b \mid c$$

$$(8) L = \{a^i b^j c^k \mid k = |i - j|\}$$

$$\begin{aligned} i > j &\rightarrow i = j + k \\ i < j &\rightarrow j = i + k \end{aligned}$$

$$S \rightarrow P \mid Q$$

$$i > j \left\{ \begin{aligned} P &= aPc \mid X \\ X &= aXb \mid \lambda \end{aligned} \right.$$

$$i < j \left\{ \begin{aligned} Q &= YZ \\ Y &= aYb \mid \lambda \\ Z &= bZc \mid \lambda \end{aligned} \right.$$

$$(9) L = \{a^i (b+c)^j d^k \mid i+k > j\}$$

$$S \rightarrow \lambda \mid APQ \mid PQD \mid APQD$$

$$\begin{aligned} P &\rightarrow aPZ \mid \lambda \\ Q &\rightarrow ZQd \mid \lambda \end{aligned} \quad \left. \vphantom{\begin{aligned} P &\rightarrow aPZ \mid \lambda \\ Q &\rightarrow ZQd \mid \lambda \end{aligned}} \right\} PQ \quad i+k > j$$

$$A \rightarrow aA \mid a$$

$$D \rightarrow dD \mid d$$

$$Z \rightarrow b \mid c$$

$$(10) L = \{a, b, c\}^* \mid N(c) = N(a) + N(b) \text{ and } N(a) \geq 1\}$$

$$S = AcAaA \mid AaAcA$$

$$A = AaAcA \mid AcAaA \mid \lambda \mid B$$

$$B = BbBcB \mid BcBbB \mid \lambda \mid A$$

Reduction:

$$S = XcXaX \mid XaXcX$$

$$X = XaXcX \mid XcXaX \mid XbXcX \mid XcXbX \mid \lambda$$